$$|r\rangle = \frac{1}{\sqrt{2}}|u\rangle + \frac{1}{\sqrt{2}}|d\rangle. \qquad (2.5)$$

There is some ambiguity in this choice, but as we will see later, it is nothing more than the ambiguity in our choice of exact directions for the *x* and *y* axes.

Next, let's look at the vector |*l*>. Here is what we know: when the spin has been prepared in the *left* configuration, the probabilities for σ_z are again equal to ½. That is not enough to determine the values $\alpha_u^* \alpha_u$ and $\alpha_d^* \alpha_d$, but there is another condition that we can infer. Earlier, I told you that |*u*> and |*d*> are orthogonal for the simple reason that, if the spin is *up*, it's definitely not *down*. But there is nothing special about *up* and *down* that is not also true of *right* and *left*. In particular, if the spin is *right*, it has zero probability of being *left*. Thus, by analogy with Eq. 2.3,

$$\langle r|l\rangle = 0$$

$$\langle l|r\rangle = 0.$$

This pretty much fixes |*l*> in the form

$$|l\rangle = \frac{1}{\sqrt{2}}|u\rangle - \frac{1}{\sqrt{2}}|d\rangle. \qquad (2.6)$$

Exercise 2.1: Prove that the vector |*r*> in Eq. 2.5 is orthogonal to vector |*l*> in Eq. 2.6.

Again, there is some ambiguity in the choice of |*l*>. This is called the *phase ambiguity*. Suppose we multiply |*l*> by any complex number *z*. That will